

Lab 1-2
Introduction to Information Security and the related tools
Preparation Lab
Installing Virtual Box and Kali Linux Operating system

1- install Virtual Box

<https://www.virtualbox.org/wiki/Downloads>

- select Windows hosts

2- Install SEED Ubuntu 20.04 VM (zip file)

Account Information of this VM

- ☐ User name: **seed**
- ☐ Password: **dees**

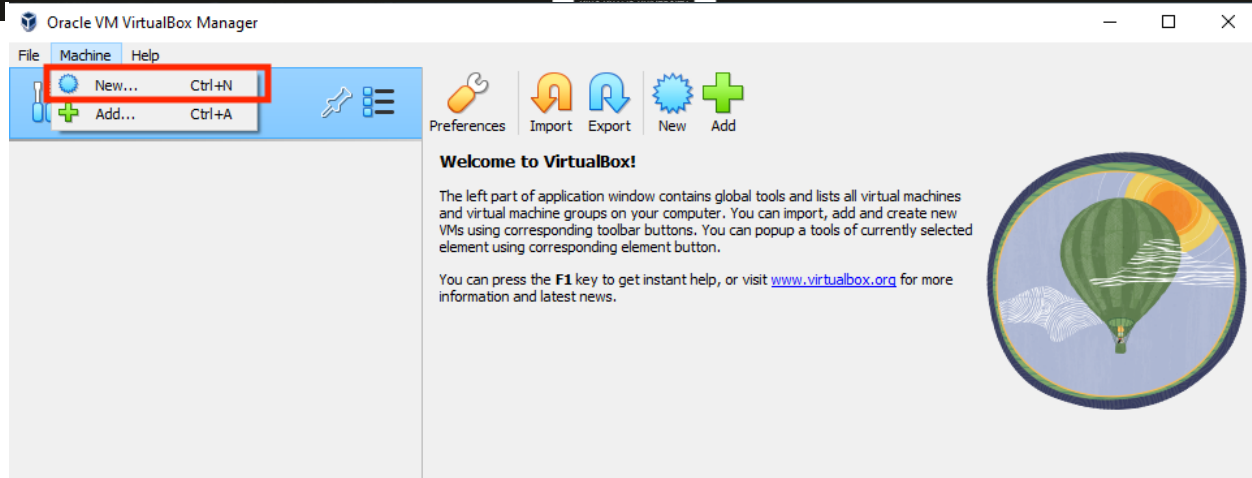
Preparation

Before installing the SEED VM, please do the following:

- ☐ Install the free [VirtualBox](#) software first. The VM has been tested on Version 6.1.16.
- ☐ Download the zip file SEED-Ubuntu20.04.zip from the [SEED website](#), unzip it, and you will get a .vdi file. This file contains the pre-built SEED Ubuntu 20.04 image. This document shows how to build a virtual machine using this image.

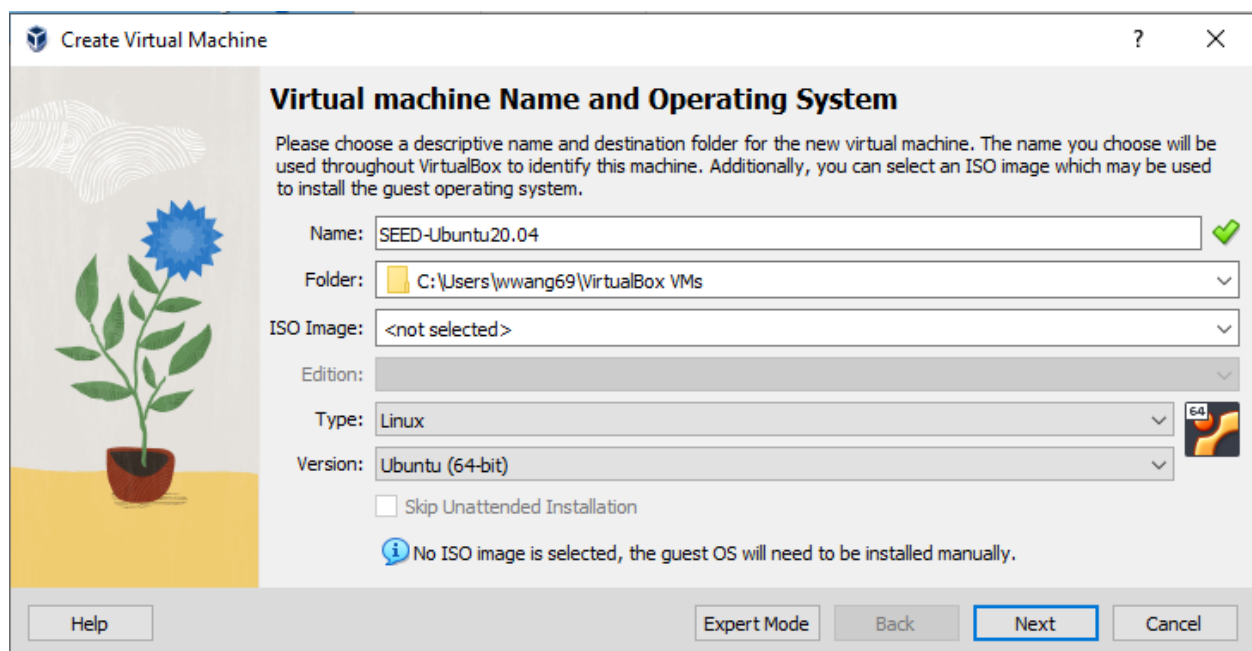
Step 1: Create a New VM in VirtualBox

We need to use New to create a new virtual machine.



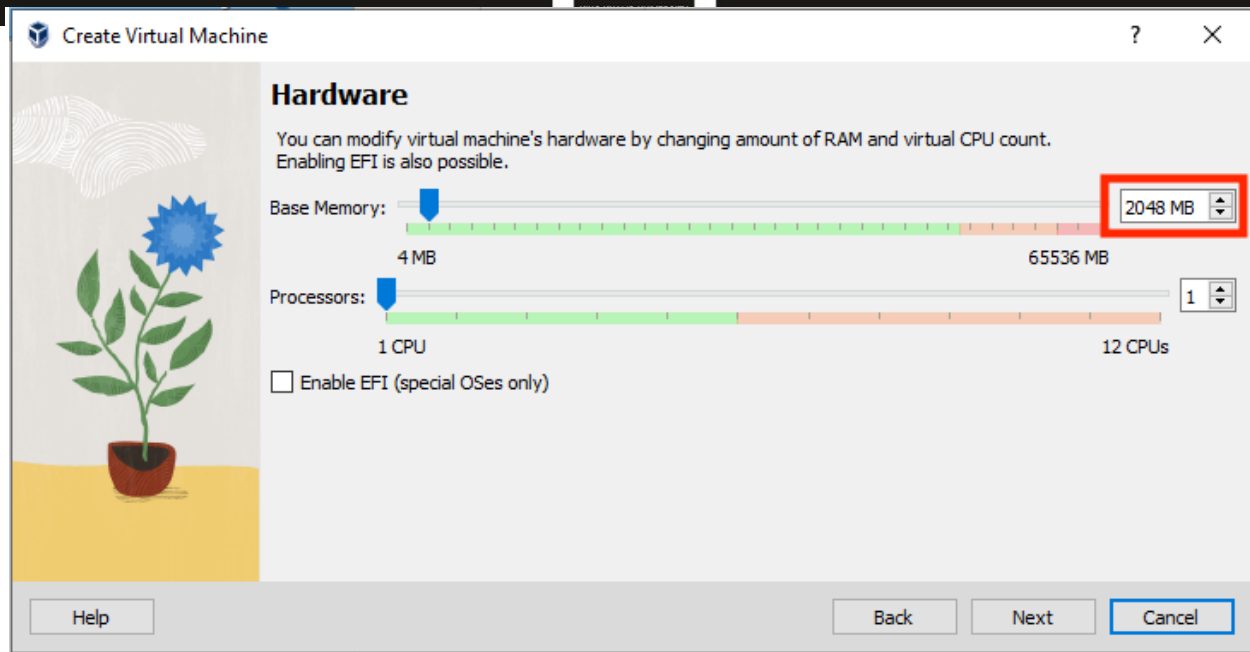
Step 2: Provide a Name and Select the OS Type and Version

Our prebuilt Ubuntu 20.04 VM is 64-bit, so pick Ubuntu (64-bit).



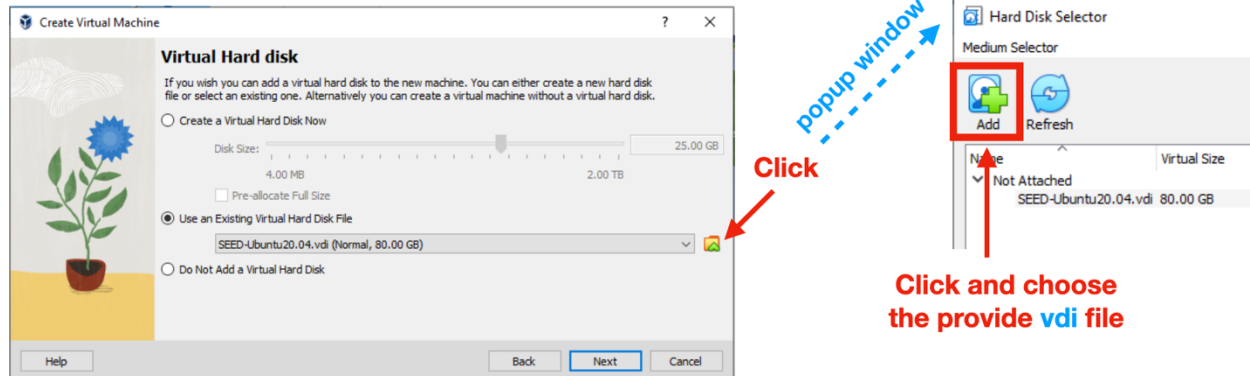
Step 3: Set the Memory Size

We need to allocate dedicated memory for the VM. 1024 MB should be sufficient, but we recommend 2GB. If your computer has more RAM, you can increase accordingly. The more memory you give to the VM, the better the performance you will get.



Step 4: Select the Pre-built VM File Provided by Us

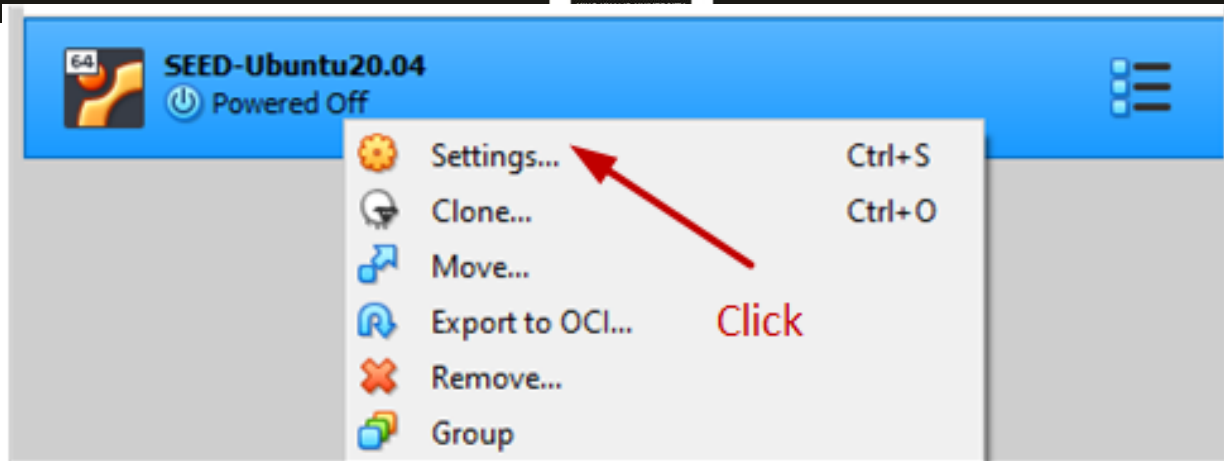
Click the folder image. On the popup window, use the Add button to select the .vdi file downloaded from the SEED website.



Note: If you get an error message saying that the UUID already exists, this is because the UUID in the selected vdi file is the same as the one used by an existing VM. You can either remove the other VM or [change the UUID](#) in the vdi file.

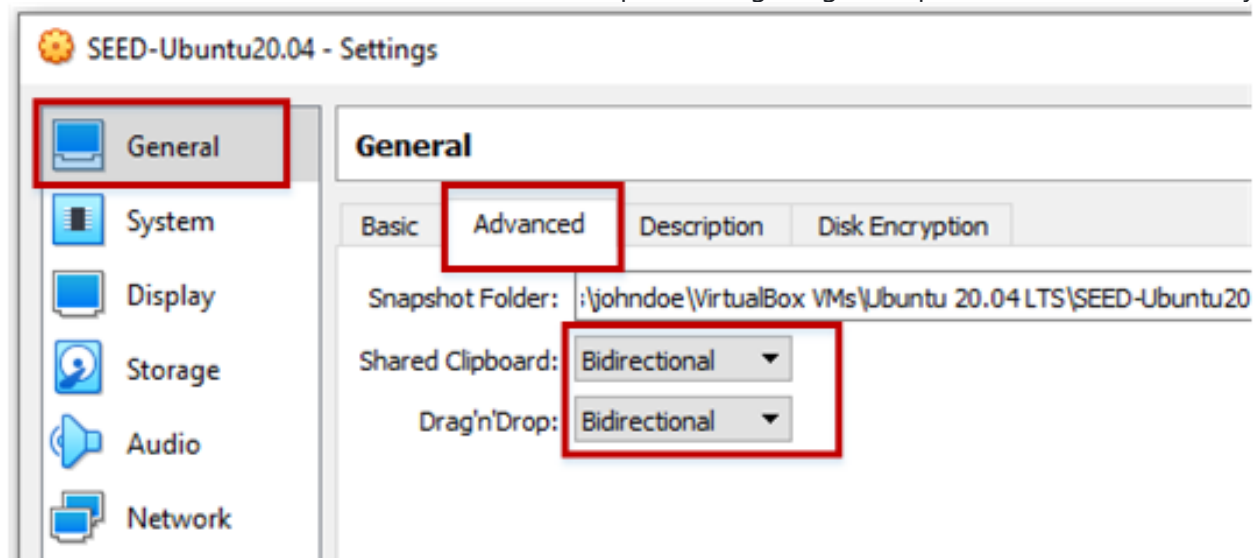
Step 5: Configure the VM

After the previous step, your VM will be created, and you will see it on VirtualBox's VM panel. We need to do some further configuration. Right-click the VM, click the Settings option, and we will see the Settings window.



Step 5.a: Enable Copy and Paste

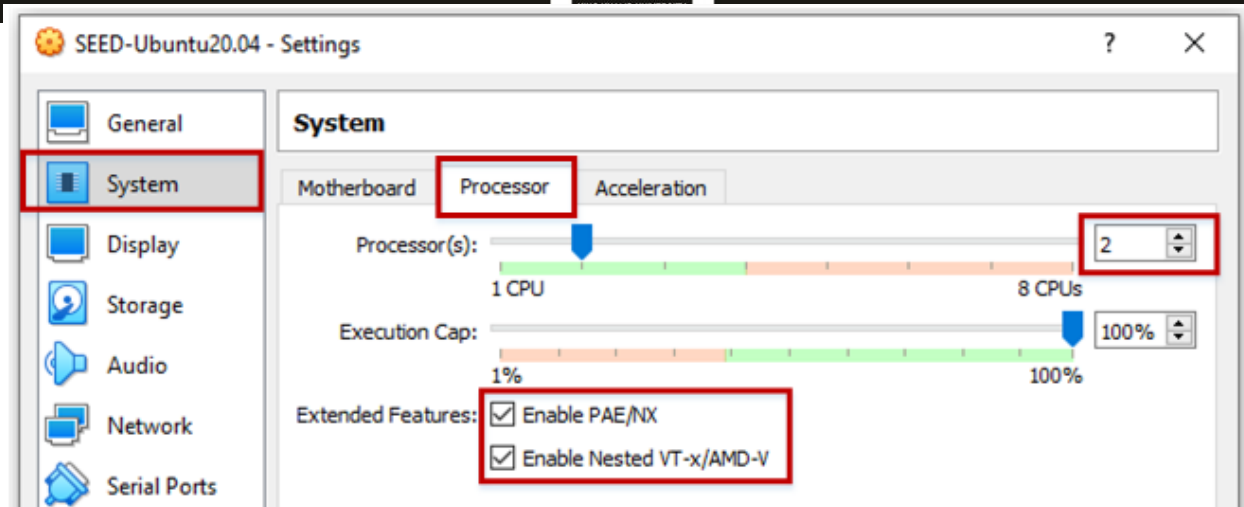
Go to the General category, and select the Advanced tab. Select Bidirectional for both items. The first item allows users to copy and paste between the VM and the host computer. The second item allows users to transfer files between the VM and the host computer using Drag'n Drop (this feature is not always reliable).



The copy-and-paste feature is very useful. If you can't do copy and paste, chances are that you forgot to do this step. You can always do it later by selecting the Devices menu item, and you will see the Shared Clipboard submenu.

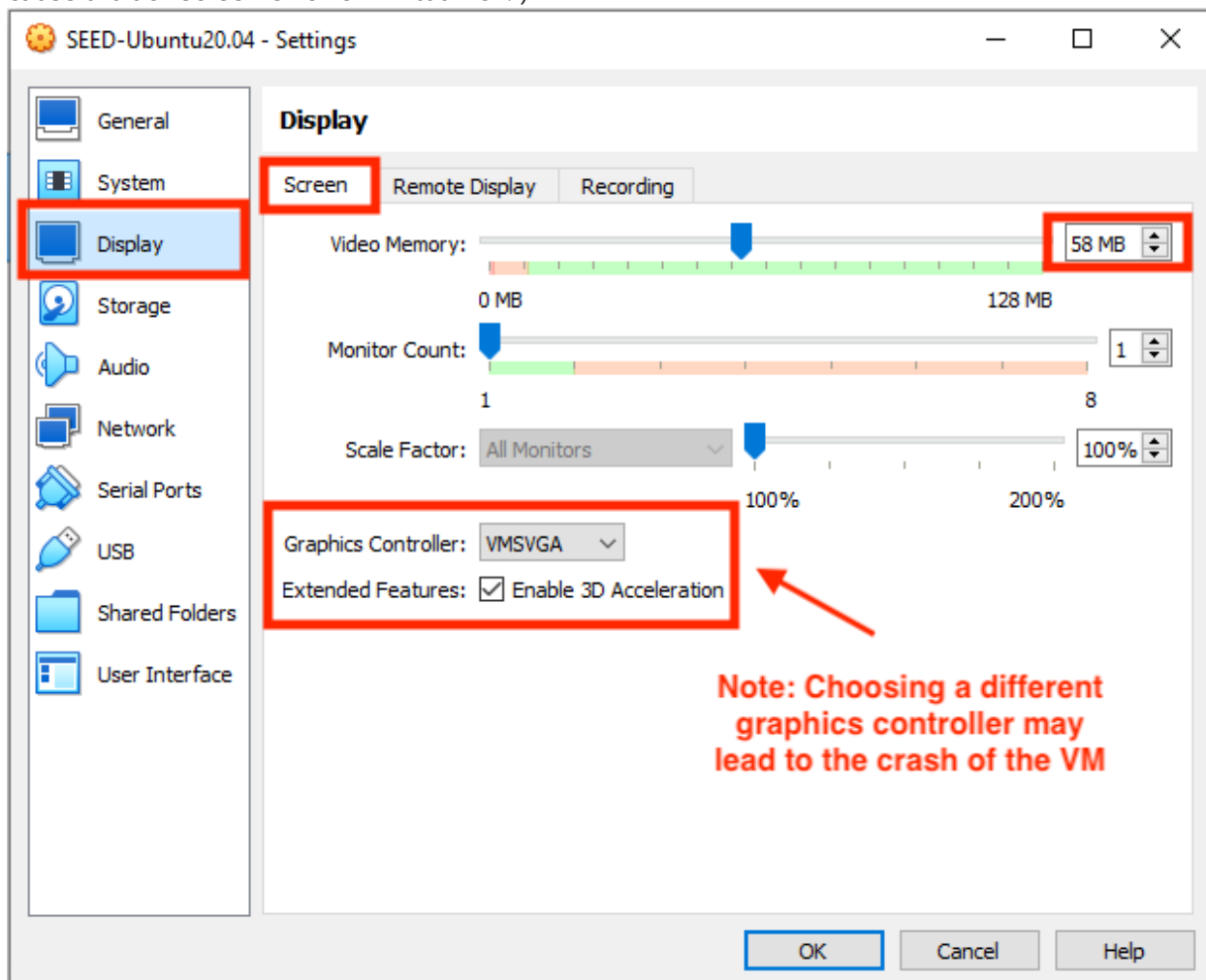
Step 5.b: CPUs

Go to the System category, and select the Processor tab. Assign number of CPUs to this VM if you prefer. Although 1 CPU may be sufficient, if the performance seems to be an issue, increase the number.



Step 5.c: Display

Go to the Display category, and select the Screen tab. If the display does not seem to work properly, try to increase the amount of video memory. In our testing, 58 MB seems to be sufficient. (Low video memory may cause a black screen error on VirtualBox.)

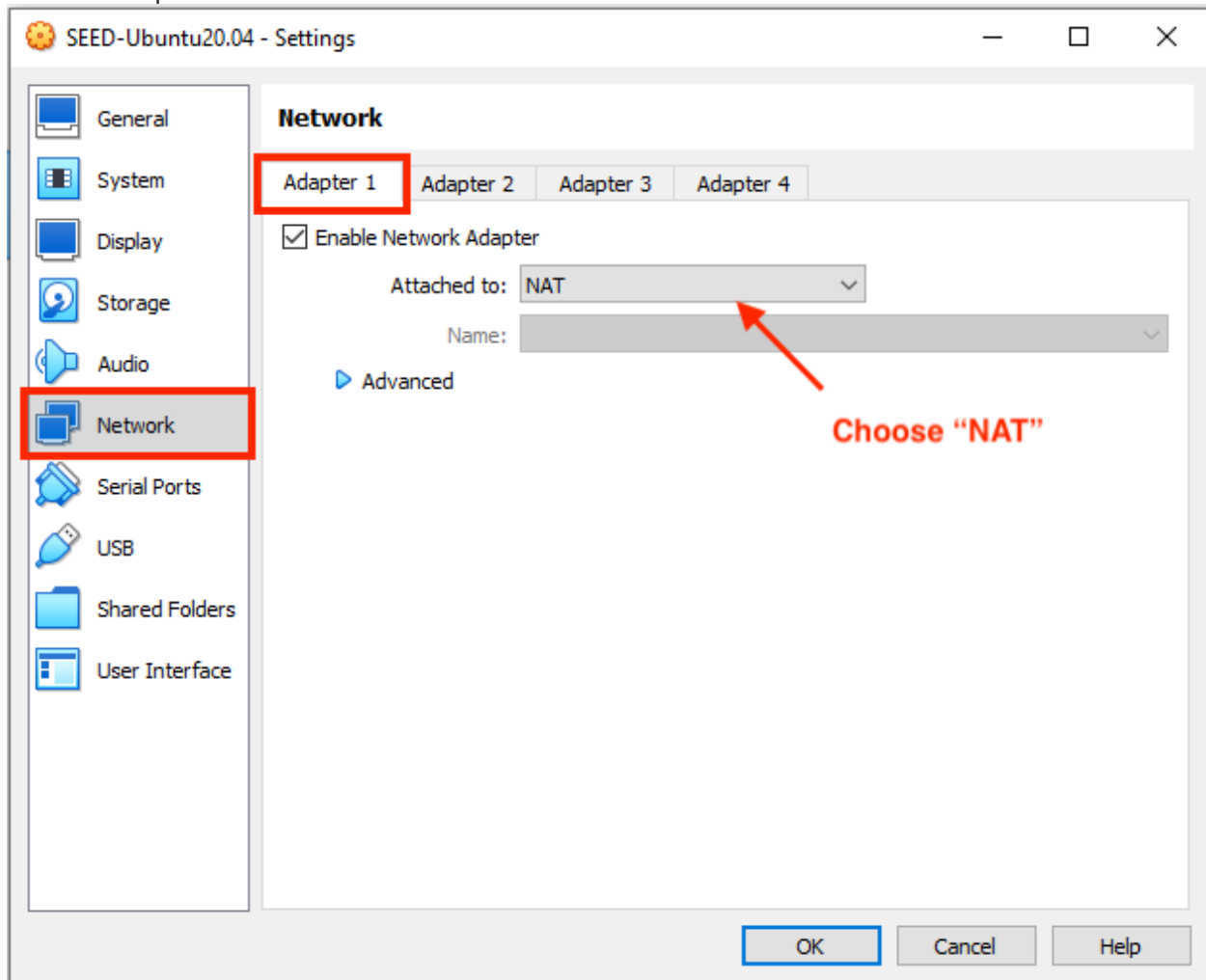


Note 1: Make sure to select VMSVGA, as choosing other graphic controllers may lead to the crash of the VM.

Note 2: If your computer's screen resolution is too high, the VM may not be able to match the high resolution. As results, your VM will be very small on your screen. To make it bigger, adjust the Scale Factor in this setting.

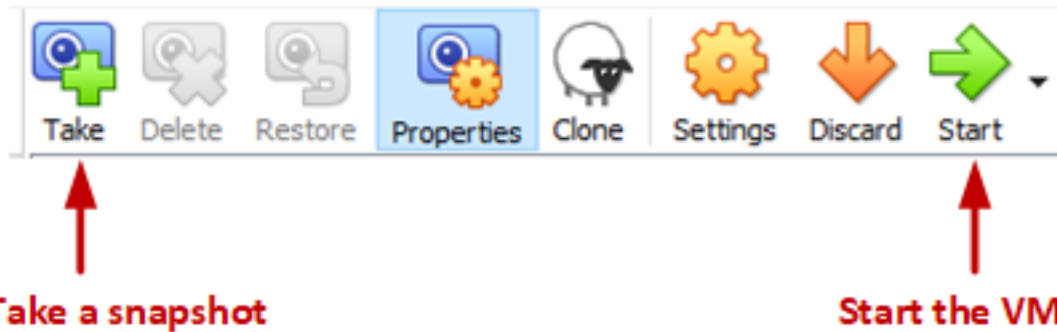
Step 5.d: Network

Go to the Network category, and select the Adapter 1 tab. We will choose the NAT adaptor.



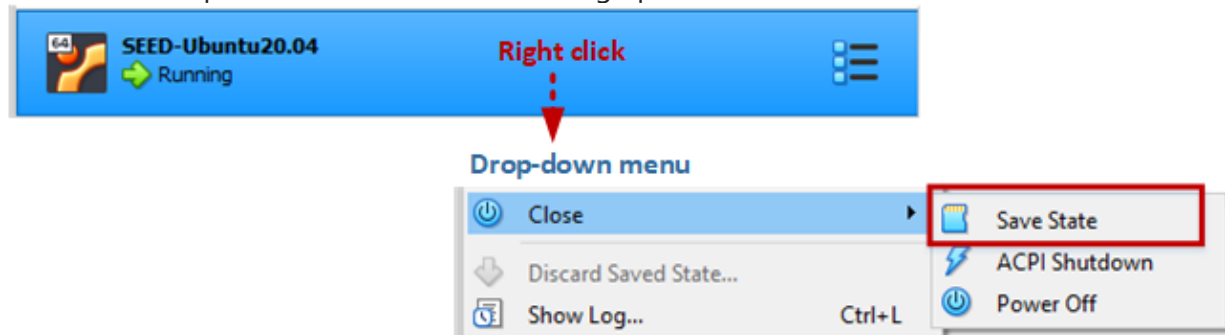
Appendix A: Start the VM and Take Snapshot

We can now start the VM. You can also use the Take button to take a snapshot of your VM. This way, if something goes wrong, you can roll back the state of your VM using the saved snapshots.



Appendix B: Stop the VM

There are many ways to stop the VM. The best way is to use the Save State. This is different from shutting down the VM. It saves the current VM state, so next time when you restart the VM, the state will be recovered. Moreover, the speed is also faster than booting up a VM.

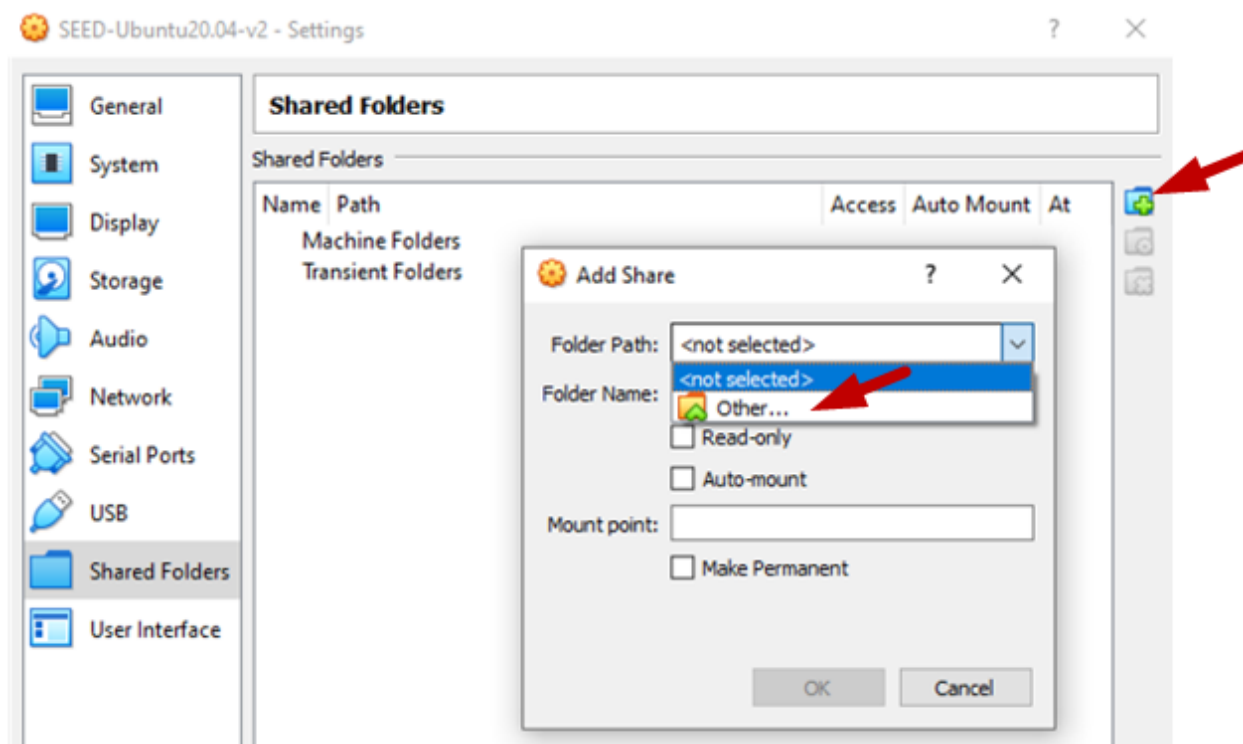
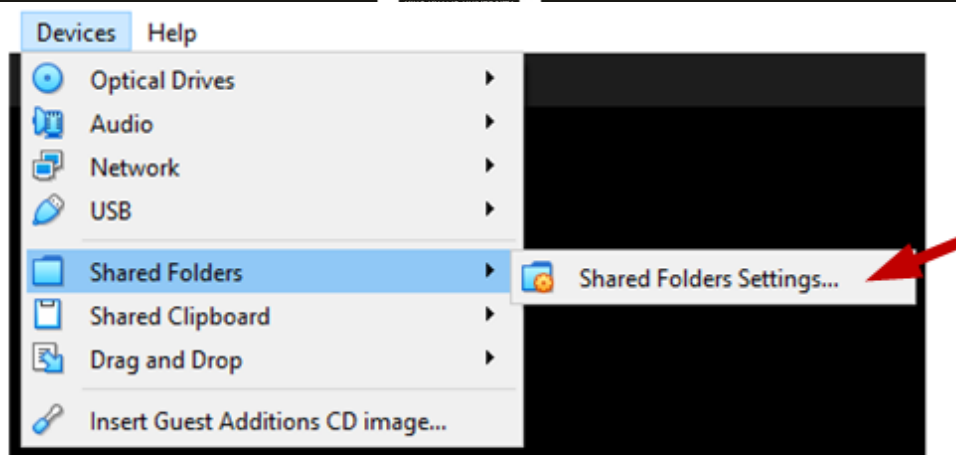


Appendix C: Creating a Shared Folder

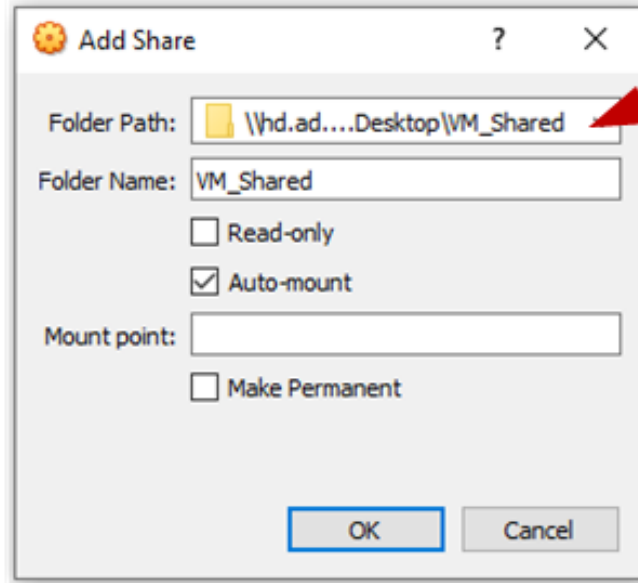
Sometimes, we need to copy files between the host machine and the VM. If you are using the VM from the cloud, you can see our cloud VM manual for instructions. Or, you can just use a cloud storage service, such as Dropbox and Google Drive to share files between your VM and host machine.

If you run the VM on your local computer, you can create a shared folder between your computer and the VM. We recommend you turn off the VM before you do Step A. Change the setting while the VM is running may crash the VM.

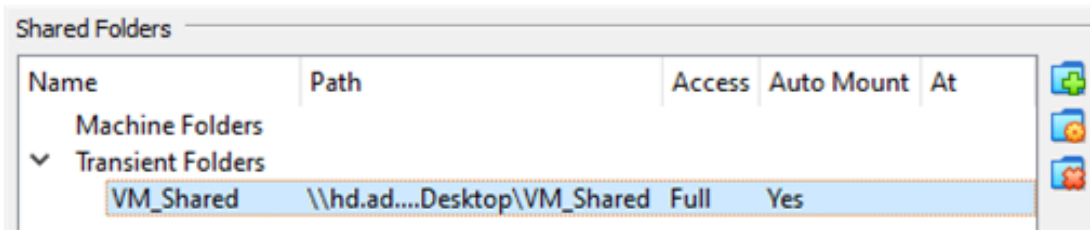
Step A. First you need to create a folder on your local computer (or using an existing folder). We will let the VirtualBox know that this folder should be shared with the VM. Go to the following menus:



Once you see a Add Share popup window, select the folder that you want to share, click OK, and you will see that the folder is now made available for sharing.



Select a folder from
your local computer



This name will be used later

Step B. Inside the VM, we need to mount the shared folder somewhere. Let's mount it to the home directory as a folder Share. We will create a folder called Share in the home directory, and then mount the shared folder VM_Shared to this Share folder using the following command. After that, you can access the shared folder from ~/Share.

```
$ mkdir -p ~/Share
$ sudo mount -t vboxsf VM_Shared ~/Share
```

On some latest version VirtualBox, the folder VM_Shared doesn't mount on automatically after the VM is restarted even if we have selected the auto-mount option. To solve this issue, we can create an auto-mount script inside the /etc/profile.d/ folder.

```
sudo vim /etc/profile.d/seedshare.sh
```

Inside the script file, add the mount command that we want to run when the system is initialized.

```
sudo mount -t vboxsf VM_Shared ~/Share
```

After the VM is started, during the login phase, all scripts under /etc/profile.d folder will be executed. This way, system automatically will mount the shared folder VM_Shared to the Share folder.

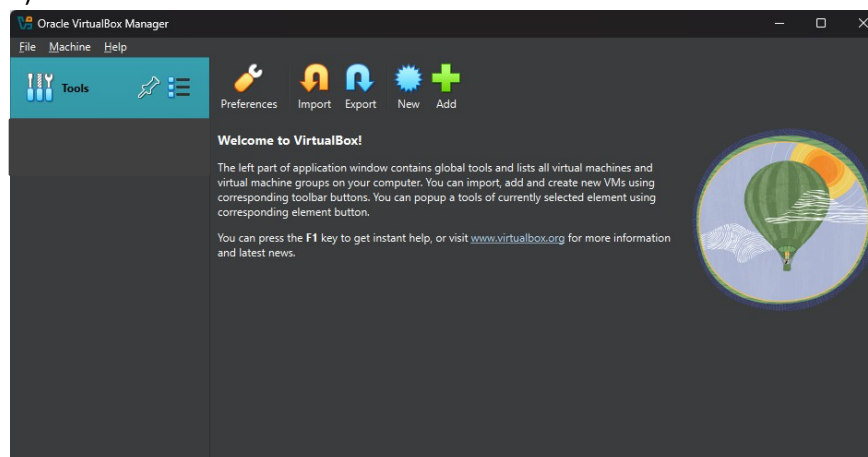
Important Note. Please only use the shared folder to copy files between the VM and the host machine, and **never use it as your working folder**. Working from the shared folder has caused many problems, especially on the permissions of the files created inside the shared folder. For example, if we unzip the Labsetup.zip file inside the shared folder, the permissions of the unzipped files will be different from those on the original files. Some labs and containers are very sensitive to those permissions.

3. Installing Kali VM on Virtual Box

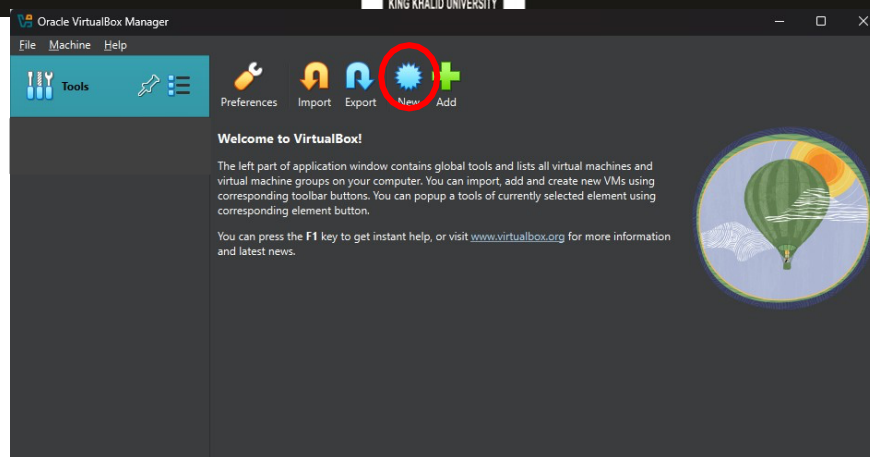
1. Install VirtualBox. The install file can be downloaded at (Already installed above)
2. <https://www.oracle.com/virtualization/technologies/vm/downloads/virtualbox-downloads.html>
3. Install Kali Linux on VirtualBox, download the Kali Linux image at <https://www.kali.org/get-kali/#kali-virtual-machines> using the VirtualBox image.
 - a. Download the VirtualBox pre-configured image. Remember where you download this to.



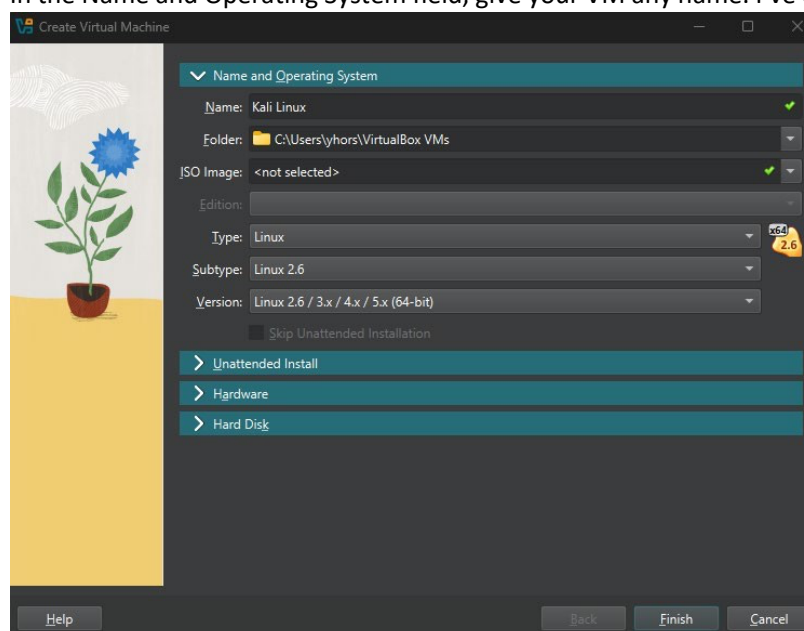
- b. In your file directory, navigate to the folder you downloaded the VirtualBox image to. Unzip the 'kali-linux-2025.1c-virtualbox-amd64' folder.
- c. Open the VirtualBox manager. It should look like this after it has been installed (see step 1).



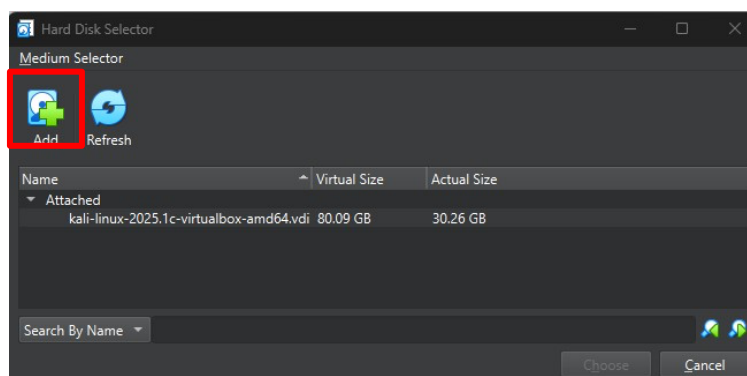
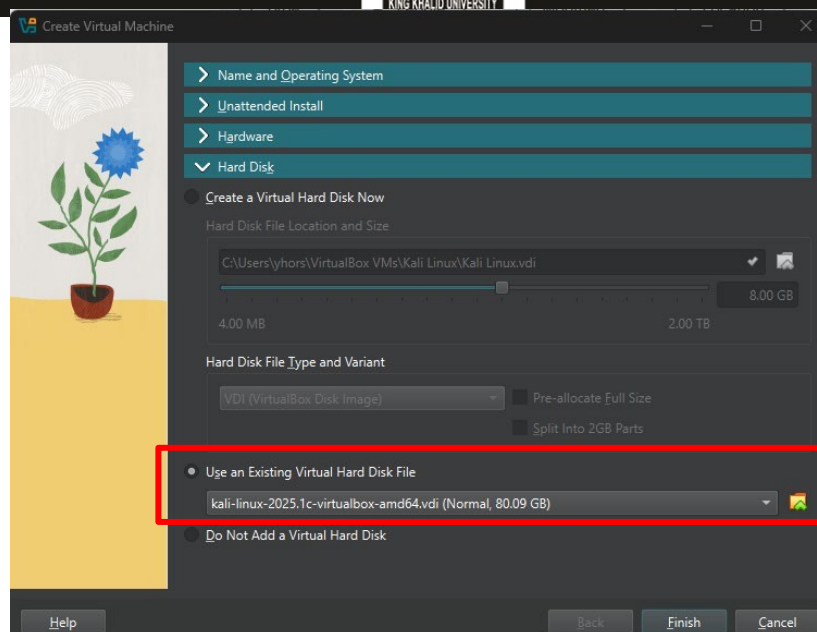
- d. Click on New



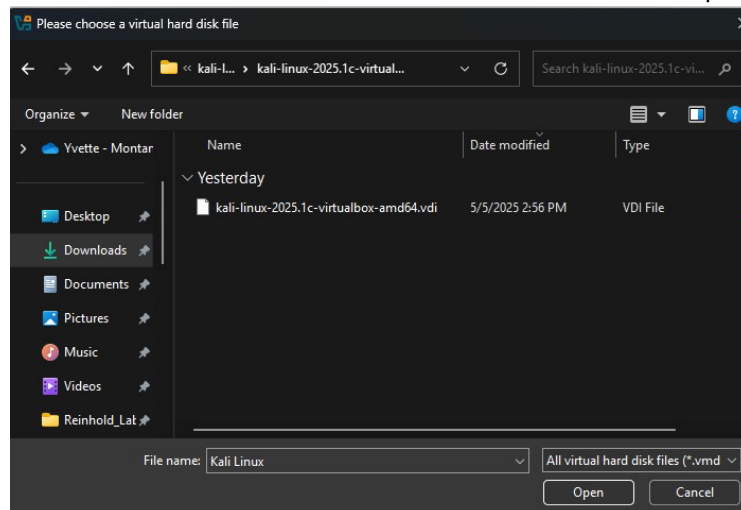
- e. In the Name and Operating System field, give your VM any name. I've called it 'Kali Linux'.



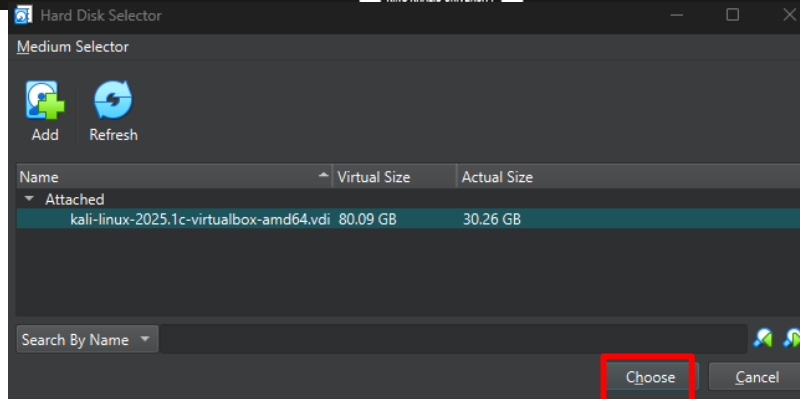
- f. In the Hard Disk field, select 'Use an Existing Virtual Hard Disk File'. Click on the file icon to the right of this field.



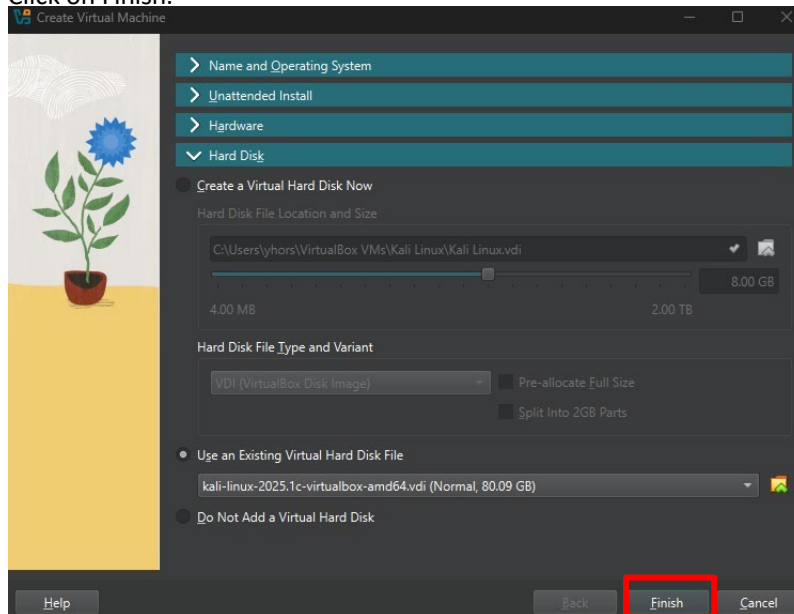
- g. In the window that open's, select Add.
- h. Navigate to the folder where you have unzipped the VirtualBox image and open it. Select the 'kali-linux-2025.1c-virtualbox-amd64.vdi' file and select open.



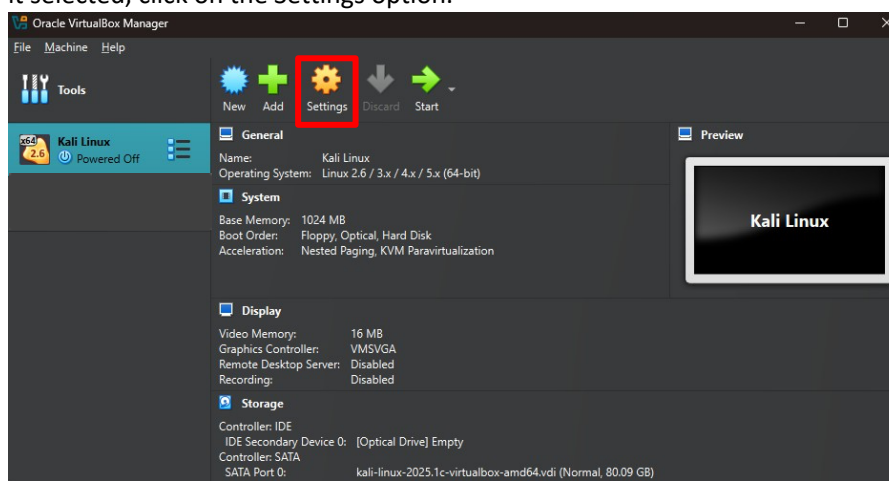
- i. Click on Choose.



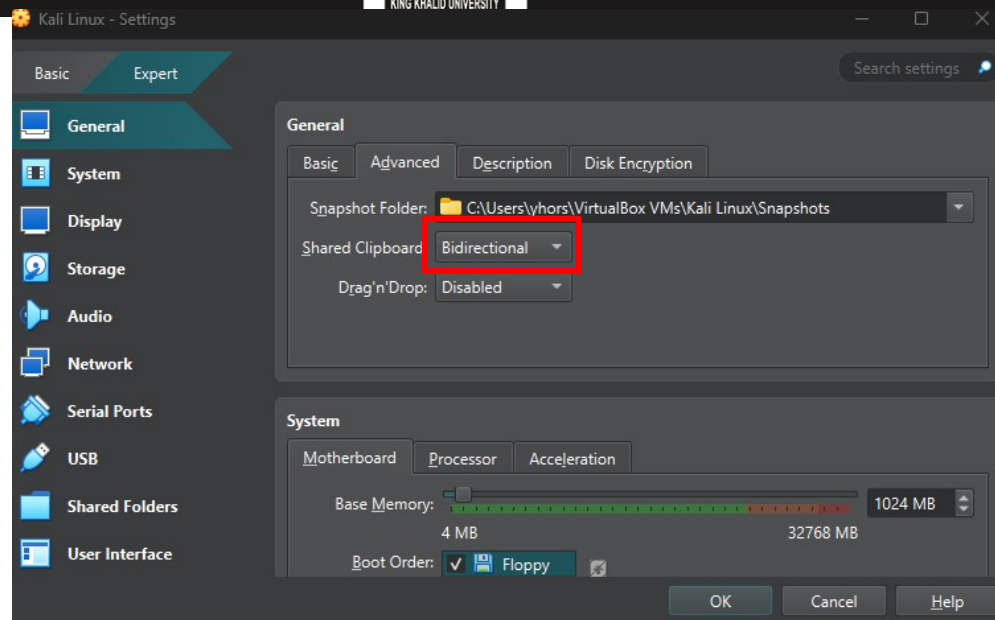
j. Click on Finish.



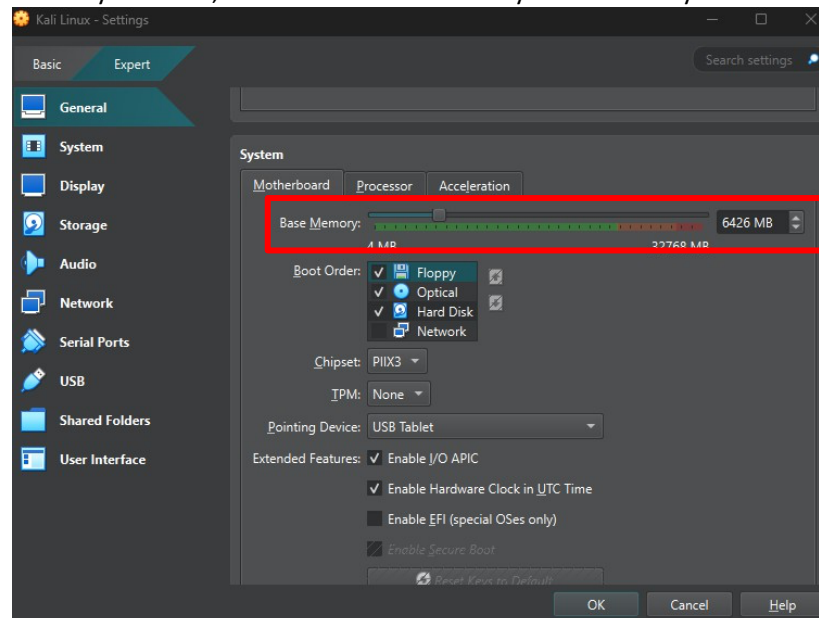
k. In the VirtualBox Manager, you will see your VM listed in the left navigation panel. With it selected, click on the Settings option.



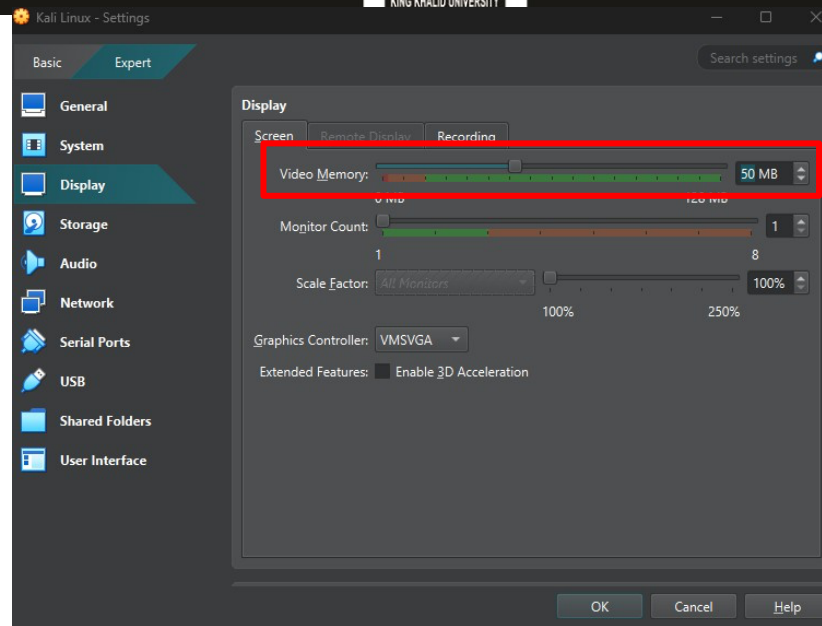
l. In the General box in the Advanced tab, set the Shared Clipboard to Bidirectional to allow you to copy and paste text from your computer to the VM, and vice versa.



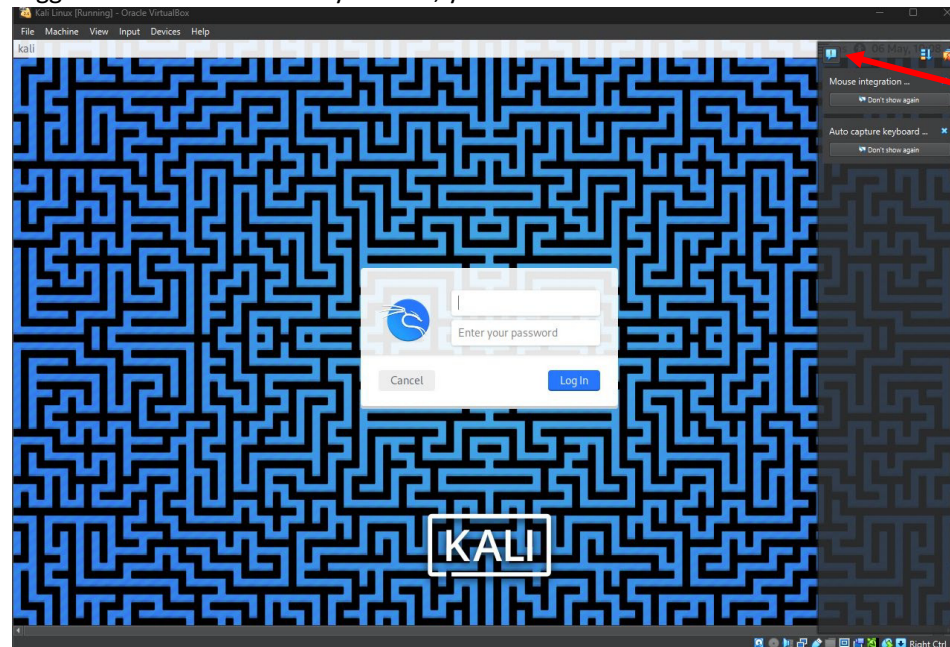
- m. In the System box, increase the Base Memory to what ever you want.



- n. In the Display box, increase the Video Memory. This will decrease lag issues while running the VM.

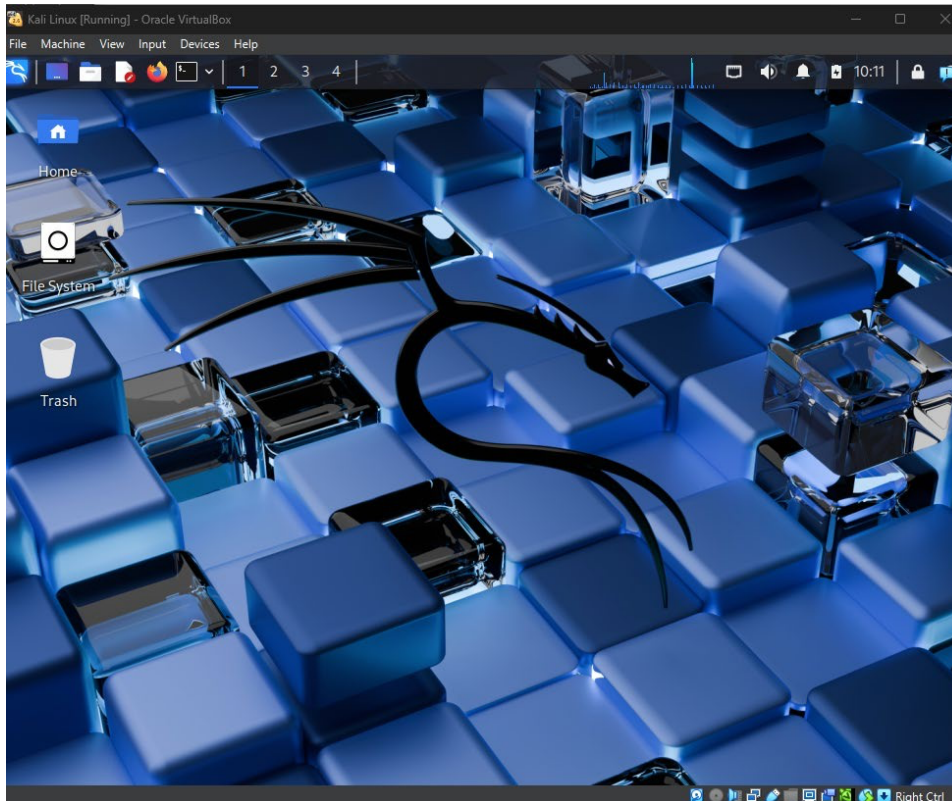


- o. Select OK to close the Settings window.
- p. Select Start In the VirtualBox Manager. While the VM is loading, you will see the screen toggle. Once the VM has fully started, you will see this screen.



You can close this window by clicking on the message icon.

- q. The username and password are preset to 'kali'.
- r. Once logged in, your VM will look like this. You are all set for the hands-on portion of the digital forensics workshop.



- s. Open the terminal. Install the required python dependencies. Run the following commands:
- ☐ `sudo apt install -y python3-dev libpython3-dev python3-pip python3-setuptools python3-wheel`
 - ☐ `python3 -m pip install -U distorm3 yara pycrypto pillow openpyxl ujson pytz ipython --break-system-packages`
- If you get an error, that is fine. The volatility3 framework will still work.
- t. Now, clone the volatility3 source code from GitHub. Make sure you run this command while in your home directory.
- ☐ `git clone https://github.com/volatilityfoundation/volatility3.git` You can verify it installed correctly with:
 - ☐ `python3 volatility3/vol.py -h`
- u. Finally, download the data and unzip the folder to the Kali Linux home directory:
https://drive.google.com/file/d/1KqKK7IGun12SeSsZfEufEhVEiptVXJsl/view?usp=drive_link;
password is 'infected'